Joshua Kofoworola

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EDUCATION

University of California, Davis - B.Eng., Computer Engineering Expected 2027

Relevant Coursework: Embedded Systems, Digital Logic, Data Structures, IoT Systems, C/C++, Python, Web Programming, Algorithm D

Berkeley City College - A.S., Computer Science GPA: 3.7

Relevant Coursework: Data Structures, Discrete Structures, Object-Oriented Programming, Introduction to Structured Programming

SKILLS / TECHNOLOGIES

Languages: Python, C/C++, JavaScript Frameworks: React Native, Node.js, Express Tools: VS Code, Git, Docker, Firebase, AWS Hardware: ESP32, Arduino, LoRaWAN, GSM Strengths: Embedded Systems, IoT, ML integration (TF-IDF, Random Forest), data pipelines, collaboration

EXPERIENCE

Software Engineer Extern - N.A.F, Kaduna, Nigeria May 2023 - Aug 2023

- Built a web-based aircraft maintenance & tracking system; improved operational visibility by 50%.
- Processed and analyzed 10,000+ JSON records (TF-IDF, feature extraction) for predictive insights.
- Trained and integrated a Random Forest model for maintenance prediction within existing workflows.
- Partnered with engineering and data teams to deploy ML outputs to dashboards and tools.

PROJECTS

Offline BLE + LoRa Chat Application Jan 2025

JavaScript, React Native, LoRa

- Built an offline chat application communicating over BLE and LoRa, working without internet or cellular infrastructure.
- Developed the mobile app and embedded firmware; prototyped and hand-assembled the hardware units.

LoRaWAN-Based Secure Communication System May 2024 - Aug 2024

C/C++, Arduino, ESP32, React Native

- Implemented an encrypted long-range IoT link using LoRaWAN on ESP32 for reliable, low-power telemetry.
- Built a React Native companion app for provisioning, diagnostics, and live graphing.

IoT Blood Sugar Monitor Oct 2024 - Dec 2024

JavaScript, Node.js, Firebase, React Native, ESP32

- Created a mobile-connected glucose monitoring pipeline with real-time Firebase sync and alert thresholds.
- Implemented BLE ingestion, user auth, device pairing, and historical trend analysis.
- Designed and built the ESP32-based measurement hardware interfacing with the mobile app.

pysearch - Full-Text Search Engine Oct 2026

Python, Flask, pytest

- Built a from-scratch full-text search engine: rule-based tokenizer, positional inverted index, TF-IDF ranking with cosine similarity.
- Supports phrase queries, boolean operators (AND/OR/NOT), and serves results via a Flask REST API with web UI; 37/37 pytest tests passing.